ORIE 5260 · MIDTERM · [GROUP NAME]

Trend following, better risk-controlled

Same 12-month momentum signal as the course baseline. A better risk model, risk caps and one trend-quality filter.

0.99 → 1.34
Sharpe ratio, 1976–2014

0.84 → 1.17
Sharpe, 2000–2014 (never used to choose)

−26% → −12%
Maximum drawdown

STARTING POINT

We first reproduced the course baseline exactly

Data

58 futures (26 commodity, 14 equity, 11 rates, 7 FX), monthly 1976–2014; daily prices for risk. A market enters once it has 12 months of signal and 36 months of vol history: 11 markets in 1976, 32 in 1990, 58 from 2010.

Discipline

Every choice made on data to 1999. 2000–2014 is reported, never used to choose. Positions at month t see data through t only; 15 tests scramble future data to prove it.

Replication vs Lecture 6

Strategy	Ours: avg / vol / SR	Course
TSMOM 12m	11.0 / 11.1 / 0.99	11.1 / 11.1 / 1.00
Static benchmark	6.9 / 14.5 / 0.48	6.9 / 14.5 / 0.47
11% vol target	11.9 / 11.1 / 1.07	11.9 / 11.0 / 1.08

Signal: sign of the sum (not compounded) of the last 12 monthly returns. Lookbacks 1/3/6m also match within 0.01.

THE UPGRADES

Four changes, each with one reason

Step	What changes	Why	SR	SR 2000+	Max DD
Course baseline	sign(12m) × 40%/vol ÷ N	Starting point	0.99	0.84	−26%
1 · Vol target 11%	Scale the book to 11% ex-ante vol (course L6)	Risk should be chosen, not an outcome	1.07	0.96	−19%
2 · Daily EWMA risk model	Vol and correlation from daily data, recent days weigh more	The forecast must react to today's market	1.28	1.18	−15%
3 · Risk caps + vol floor	≤10% of risk per market, ≤40% per asset class	No market or class should dominate	1.29	1.16	−14%
4 · Trend-continuity filter	Halve trends built on a few jumps	Steady trends persist, jumps fade	**1.34**	**1.17**	**−12%**

Every row trades the same markets in the same months, before transaction costs. Step 2 carries most of the gain.

UPGRADE 1 · VOL TARGETING

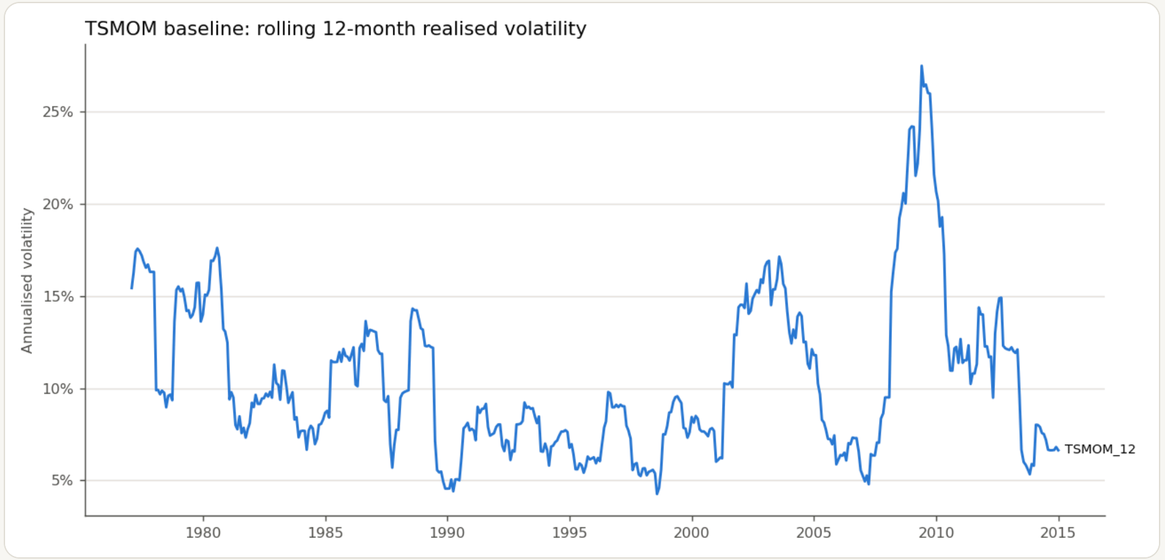
Risk should be a decision, not an outcome

Problem. Each position is sign × 40%/vol ÷ N. The 40% is an arbitrary constant and nothing controls the total, so the book's risk depends on how many markets trade and how correlated they are.

Evidence. Realised volatility swung from 5% to 27% (2009).

Fix. Every month, scale the whole book so its forecast volatility is 11%: bigger when markets are calm, smaller when they are not.

Sharpe 0.99 → 1.07, drawdown −26% → −19%.



UPGRADE 2 · DAILY EWMA RISK MODEL

The risk forecast must react to today's market

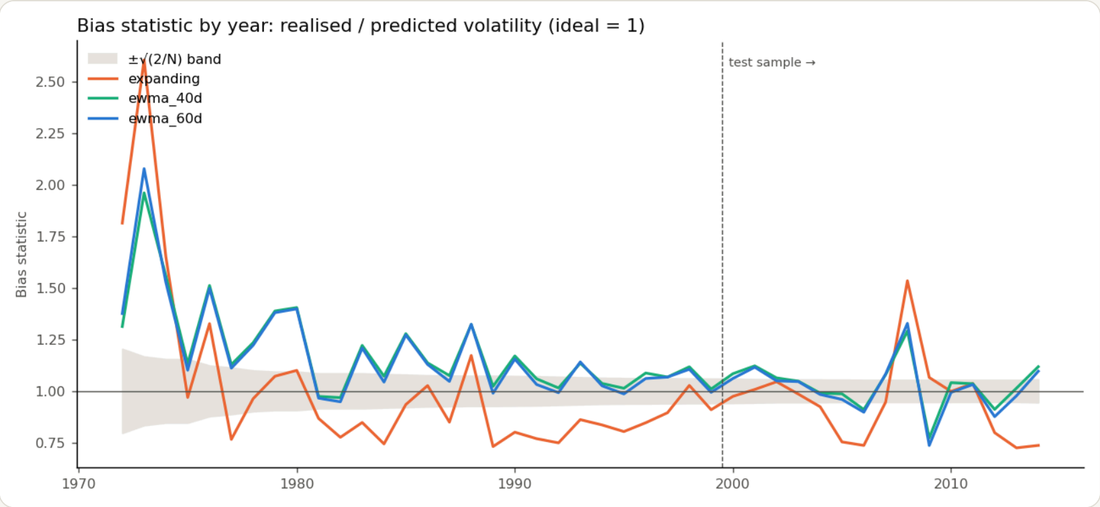
Too slow. The course estimate weights 1975 as much as last month: in 2008 realised risk was 1.5× the forecast, in calm years 0.75×.

Fix. Daily returns, recent days weigh more: vol half-life 60 days; correlation half-life 120 days on 3-day returns, because markets close hours apart (S&P–Nikkei 0.19 daily, 0.54 over 3 days).

Correlation matters. Assuming zero correlation understates the TSMOM book's risk about 2×.

Half-lives chosen on data to 1999 by the monthly forecast error (QLIKE), not by Sharpe.

Sharpe 1.07 → 1.28; from 2000: 0.96 → 1.18.



How much did the risk forecast deviate each year? Realised ÷ forecast vol; 1 = no deviation. Orange: course estimate. Blue: EWMA.

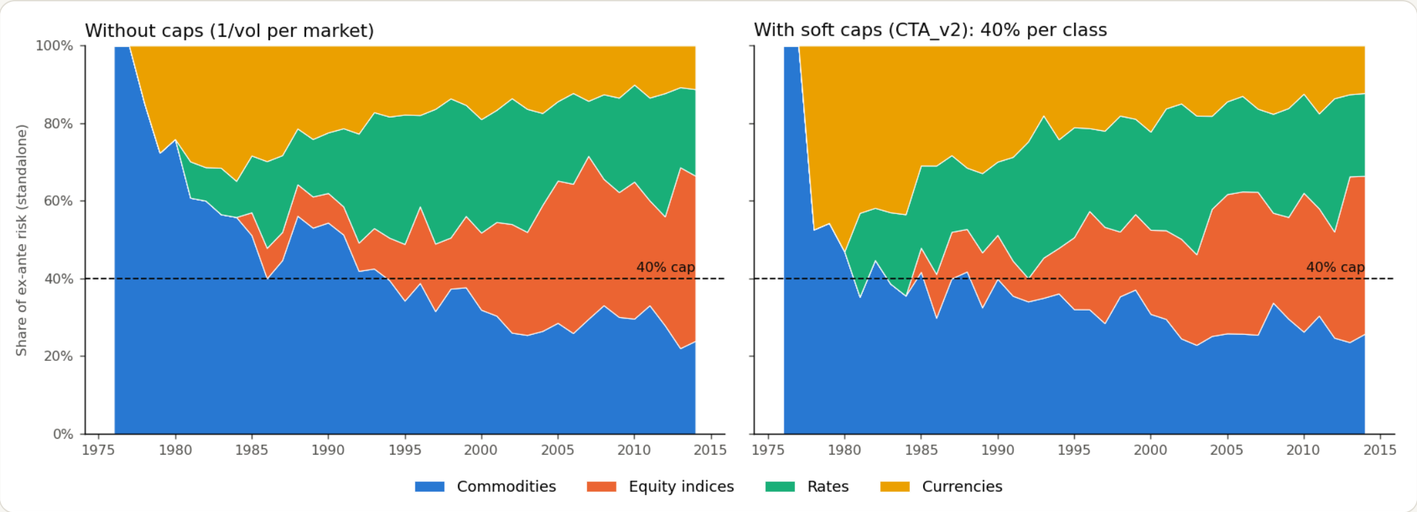
UPGRADE 3 · RISK CAPS AND A VOL FLOOR

No single market or asset class should run the book

Why caps. Commodity futures listed first, so for years the book was mostly commodities, not because their signals were stronger.

Soft caps: 10% of risk per market, 40% per class; share = cap × tanh(λ × raw / cap). Near the cap, each step needs a stronger signal.

Vol floor. At zero rates Eurodollar's vol fell to 0.07% a year, so 1/vol sizing meant 14× notional and 31× gross. Floor: vol ≥ 50% of the market's own history. Peak gross 31× → 12.5×, Sharpe unchanged.



Commodities' average risk share: 45% → 37%. The cap binds mostly before 1992. In 1976–79 only one or two classes trade, so no cap can apply.

UPGRADE 4 · TREND-CONTINUITY FILTER

Steady trends persist; jump-driven trends fade

Measure. Continuity = sign(12m return) × (share of up days − share of down days) over the past year: the "frog in the pan" idea (Information Discreteness, Da, Gurun & Warachka 2014).

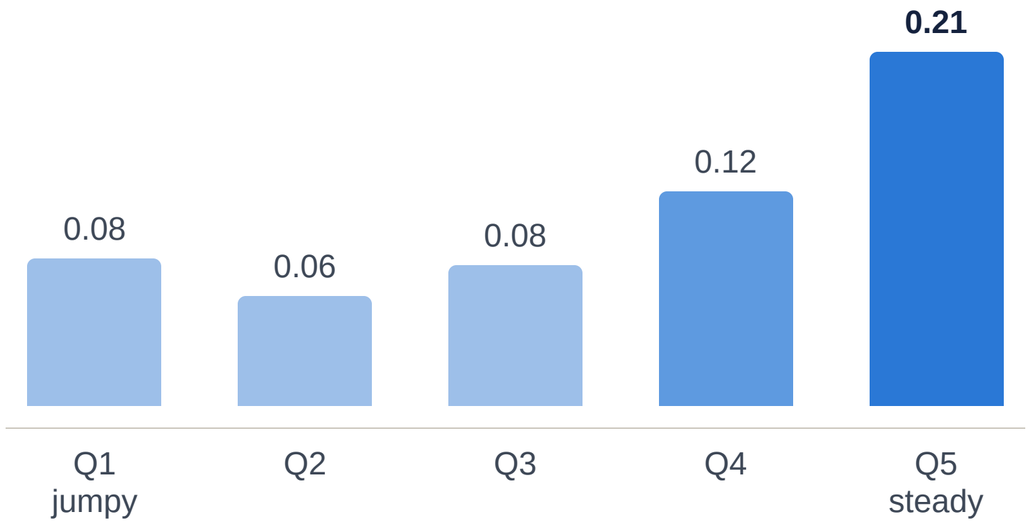
Rule. Keep the 12-month direction. Rank markets by continuity each month; the lower half gets half a position. The vol target then rescales the book, so risk shifts toward steady trends.

Result. Sharpe to 1999: 1.36 → 1.44. From 2000: 1.16 → 1.17. Drawdown −14% → −12%, no extra turnover.

Honest read: consistent in both periods, but the gain after 2000 is within noise.

Next-month payoff of a TSMOM position, by continuity group (data to 1999)

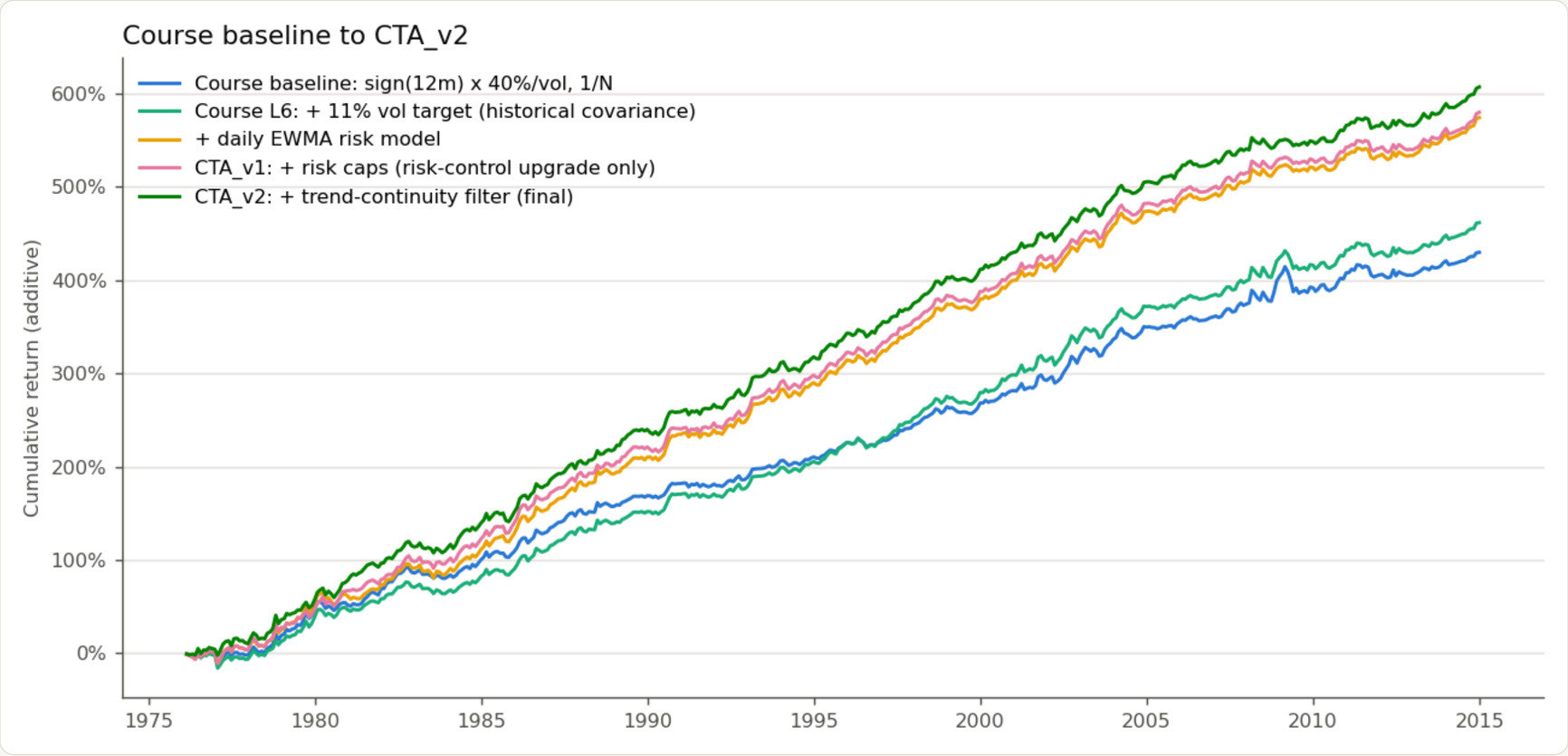
Each month, markets are sorted by continuity into 5 equal groups: Q1 = most jump-driven, Q5 = steadiest. Bar = average return per unit of risk.



Q5 − Q1 = 0.13 per month, t = 2.6

RESULTS

Same signal, same risk level, more return



8 · Monthly returns added up, 1976–2014, before costs. All versions run at about 11% volatility.

WHAT WE TESTED AND DROPPED

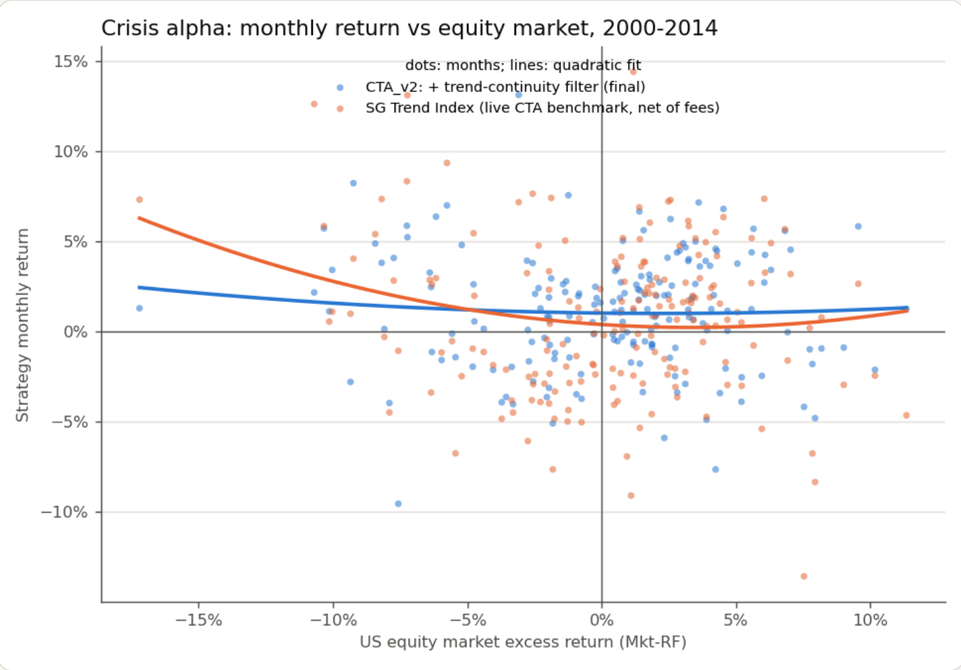
The 12-month signal is hard to beat out of sample

Idea (on the CTA_v1 engine)	SR to 1999	SR from 2000	Why not adopted
Reference: sign(12m)	1.36	1.16	
Multi-horizon signs, 1/3/12m	1.46	1.13	Gain vanishes after 2000; 2.3× turnover
Continuous signal (return ÷ its std error)	1.35	1.18	No better before 2000; drawdown −23%
Risk-adjusted 12m (each month ÷ its vol)	1.37	1.19	+0.02 / +0.03: within noise
Regime filter p_trend, best of 18 settings	1.37	1.15	A hard 40% cut: 1.13 / 1.03; removes diversification

Pattern: faster signals helped before 2000 and faded after. The same holds in a century of data (Hurst, Ooi & Pedersen 2017), so we kept the simple signal.

IS IT STILL TREND FOLLOWING? · 2000–2014

Yes, but less convex in crises than a live CTA



2000–2014	CTA_v2	SG Trend
Sharpe	1.17	0.53
Correlation with each other	0.63	–
Worst 10% equity months	+1.9%	+3.6%
Mkt² coefficient (smile)	0.38	1.43

SG Trend is net of fees; ours is before costs.

Next. Transaction costs (Lecture 8). Restore crisis convexity: let the vol target cut less when trends are strong in a sell-off. Then revisit faster signals with costs included.